www.csvtuonline.com

Roll No. 3113713076

337355 (37)

B.E. (Third Semester)
Examination, April/May 2015
(New Course)
(Branch : Mech.)
Engineering Thermodynamics

Time : Three Hours] [*Maximum Marks : 80*
[*Minimum Pass Marks : 28*

Note : Part (a) is compulsory. Solve any two ques. from (b), (c) and (d). Use of steam table and mollier chart is permitted.

UNIT-1

1. (a) Define : 2

(i) Heat engine

(ii) Heat pump

(iii) Refrigerator

(b) m_1 kg of water at T_1 is isobarically and adiabatically mixed with m_2 kg of water at T_2 ($T_1 > T_2$). Show

www.csvtuonline.com

www.csvtuonline.com

that for equal masses of water, the entropy change of the mixture is given by : 7

$$(dS)_{\text{universe}} = 2mc_p \log_e \left[\frac{T_1 + T_2}{2\sqrt{T_1 T_2}}\right]$$

(c) One kg of ice at – 4°C is exposed to the atomsphere which is at 20°C. The ice melts and comes into equilibrium with the atmosphere. Determine the entropy increase of the uninverse. For ice, take c_p = 2·09 kJ/kgK and latent heat of fusion = 333 kJ/kg. Proceed to calculate the minimum amount of work necessary to convert the water back into ice out – 4°C. 7

(d) A reversible heat engine operates between two reservoirs at the temperature of 600°C and 40°C. The engine drives a reversible refrigerator which operates beween reservoirs at temperature of 40°C and – 20°C. The heat transfer to the heat engine is 2000 kJ and the net work output of the combined engine refrigerator plant is 360 kJ. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(i) Evaluate the heat transfer to the refrigerant and the net heat transfer to the reservoir at 40°C. 7

(ii) Reconsider (i) given that the efficiency of the heat engine and the COP of the refrigerator are each 40% of their maximum possible value.

UNIT-2

2. (a) Define dead state. 2

(b) Derive the following relation for the difference in heat cerpacities :

$$C_P - C_V = \frac{\beta^2}{\alpha} T.V.$$

where α and β are isothermal compressibility and volume expansivity respectively. The other symbols have their visual meanings. 7

(c) Set up the TdS equation in the following form : 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(i) $TdS = C_V\, dT + \left(\frac{\beta T}{\alpha}\right) dV$

(i) $TdS = C_P dT - \beta u \cdot T dP$

where α and β are isothermal compressibility and volume expansivity respectively. The other symbols have their visual meaning.

(d) Air in a cylinder containing a piston is heated at constant pressure by addition of 100 kJ/kg of air. The initial temp of air is 28°C. The temperature of surroundings is 21°C. Determine the change in arravilability per kg of air. Take, $C_p = 1{\cdot}005$ kJ/kgK and $p_0 = 1$ bar.

UNIT-3

3. (a) Define : 2

(i) Law of corresponding states

(ii) Compressibility factor

(b) Determine the value of constants 'a' and 'b' from van der Waals, equation of state. 7

www.csvtuonline.com

www.csvtuonline.com

(c) Develop the following form of van der Waals' equation in terms of reduced parameter : 7

$$\left(P_r + \left(P_r + \frac{3}{v_r^2}\right)(3v_r - 1) = 8T_r.\right.$$

(d) A vessel contains 10 kg of oxygen, 8 kg of nitrogen and 25 kg of carbondioxide at 375 K temperature and 250 K Pa pressure Make calculations for the capacity of the vessel, the partial pressure of each gas present in the vessel, and the total pressure in the vessel when the temperature is raised to 450 K. 7

UNIT-4

4. (a) Define pure substance. 2

(b) Explain the process of steam generation along with its Temp. V_s Heat input diagram. 7

(c) Steam initially at 0·3 MPa, 250°C is cooled at constant volume : 7

(i) At what temperature will the steam becomes saturated vapour.

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(ii) What is the quality at 80°C ? What is the heat transferred per kg of steam in cooling from 250°C to 80°C ?

(d) 5 kg of steam at 10 bar 0·9 dry is heated at constant temperature and the final pressure of steam is 4 bar. Find the condition of steam after heating, change in internal energy, heat transfer and work done during the process.

UNIT-5

(a) Define boiler efficiency. 2

(b) Explain the working and construction of Cochran boiler with heat labelled sketch. 7

(c) Explain draught and its classification. 7

(d) Explain : 7

(i) Water level indicator

(ii) Pressure guage

(iii) Superheater

(iv) Economiser

www.csvtuonline.com